

Week 7 tutorial: reasoning with uncertainty

Base rates, Bayes nets, belief updates, and calibrated confidence

COMP9414 Artificial Intelligence

Week 7

What usually goes wrong?

Common mistakes

- ▶ confusing $P(A | B)$ with $P(B | A)$
- ▶ ignoring the base rate
- ▶ treating likelihood rows as belief distributions
- ▶ assuming different sensors are independent
- ▶ treating Bayes-net arrows as automatically causal
- ▶ choosing the most likely state without considering cost
- ▶ treating a neural-network score as a calibrated probability

Question: Which of these mistakes is most tempting?

Two friends fail the diagnostic

Two friends take beginning-of-term diagnostic quizzes.

- ▶ Alex takes the easy class.
- ▶ Bea takes the hard class.
- ▶ Both fail the diagnostic.
- ▶ The diagnostic has the same quality in both classes.

$$P(\text{Fail} \mid \text{AtRisk}) = 0.90 \quad P(\text{Fail} \mid \text{NotAtRisk}) = 0.20$$

$$P(\text{AtRisk} \mid \text{EasyClass}) = 0.05 \quad P(\text{AtRisk} \mid \text{HardClass}) = 0.40$$

Question: Both students failed. Who is more likely to be genuinely at risk?

Same evidence, different prior

$$P(\text{AtRisk} \mid \text{Fail, Easy}) = \frac{0.90 \cdot 0.05}{0.90 \cdot 0.05 + 0.20 \cdot 0.95} = \frac{0.045}{0.235} \approx 0.19$$

$$P(\text{AtRisk} \mid \text{Fail, Hard}) = \frac{0.90 \cdot 0.40}{0.90 \cdot 0.40 + 0.20 \cdot 0.60} = \frac{0.360}{0.480} = 0.75$$

Student	Class	Result	Posterior risk
Alex	easy	failed	19%
Bea	hard	failed	75%

Takeaway: The diagnostic result matters, but it is interpreted against a prior.

Natural frequencies before formulas

Easy class: 1000 students

Group	No.	Fail rate	Fail diag.
at risk	50	90%	45
not at risk	950	20%	190
total failed			235

$$45/235 \approx 0.19$$

Hard class: 1000 students

Group	No.	Fail rate	Fail diag.
at risk	400	90%	360
not at risk	600	20%	120
total failed			480

$$360/480 = 0.75$$

Question: Which presentation makes the base-rate effect easier to see?

Formulas or frequencies?

Likelihood row versus belief row

Five-cell corridor. Door cells are 2 and 4.

Prior: $[0.2, 0.2, 0.2, 0.2, 0.2]$

Likelihood after seeing a door: $[0.2, 0.8, 0.2, 0.8, 0.2]$

Unnormalised product: $[0.04, 0.16, 0.04, 0.16, 0.04]$ $Z = 0.44$

Posterior: $[0.091, 0.364, 0.091, 0.364, 0.091]$

Warning: The likelihood row is not a belief distribution. It does not need to sum to 1.

Question: Which rows must sum to one?

Likelihood row versus belief row: answer

Prior and posterior are belief distributions, so they must sum to one.

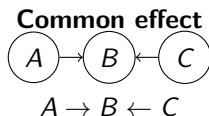
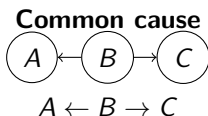
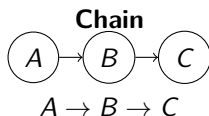
$$\sum_x bel(x) = 1 \quad \sum_x bel'(x) = 1$$

Likelihoods are different:

$$P(\text{observation} \mid x)$$

They are probabilities of the observation under each possible state, not a probability distribution over states. Unnormalised products also do not need to sum to one; dividing by Z makes the new belief sum to one.

Three graph patterns



Pattern	Middle unknown	Middle known
chain	path can carry information	path is blocked
common cause	path can carry information	path is blocked
common effect	path is blocked	path can open

Warning: The common effect case behaves differently.

Explaining away with numbers

Let B = BatteryLow, M = MotorFault, and S = MovingSlowly.

$$P(B) = 0.1, \quad P(M) = 0.2$$

B	M	$P(S = \text{true} \mid B, M)$
F	F	0.05
T	F	0.40
F	T	0.70
T	T	0.95

$$P(M \mid S) = \frac{0.2(0.95 \cdot 0.1 + 0.70 \cdot 0.9)}{0.213} = \frac{0.145}{0.213} \approx 0.68$$

$$P(M \mid S, B = \text{true}) = \frac{0.95 \cdot 0.2}{0.95 \cdot 0.2 + 0.40 \cdot 0.8} \approx 0.37$$

Takeaway: Once low battery is known, motor fault becomes less necessary as an explanation.

Bayes filter trace

Start after seeing a door in the five-cell corridor:

$$bel = [0.091, 0.364, 0.091, 0.364, 0.091]$$

Motion model for moving right:

$$P(\text{success}) = 0.8, \quad P(\text{stay}) = 0.1, \quad P(\text{overshoot}) = 0.1$$

Before calculating:

- ▶ Where should the belief move?
- ▶ Should it become sharper or more spread out?

Bayes filter trace: answer

The motion-updated belief is:

[0.009, 0.109, 0.309, 0.146, 0.427]

Boundary assumption: probability mass that would move beyond cell 5 remains in cell 5.

Takeaway: Sensing tends to sharpen belief. Uncertain motion tends to shift and spread belief.

Most likely is not the same as safe

Current belief after moving right:

[0.009, 0.109, 0.309, 0.146, 0.427]

The most likely cell is cell 5.

Question: Should the robot act as if it is definitely in cell 5?

- A. Yes, because cell 5 has the highest probability.
- B. No, because the robot is still uncertain.
- C. It depends on the cost of being wrong.
- D. It depends on whether more evidence is available.

Most likely is not the same as safe: answer

C and D are the most complete answers.

A maximum-probability state can be useful, but the full belief distribution still matters.

If being wrong is cheap, acting on the most likely cell may be acceptable.
If being wrong is expensive, the robot should consider risk, utility, and whether another observation is available.

Thresholds from consequences

A robot can pick up an object or ask for help. Let

$$p = P(\text{object is safe}).$$

Action	safe object	fragile object
pick up	+10	-100
ask human	+2	+2

$$EU(\text{PickUp}) = 10p - 100(1 - p) \quad EU(\text{AskHuman}) = 2$$

Pick up only if:

$$10p - 100(1 - p) > 2 \implies 110p > 102 \implies p > 0.927.$$

Takeaway: A 0.9 confidence score can be too low if the downside is large.

Question: What happens to the threshold if asking a human becomes more expensive?

Calibration by bins

Bin	Count	Avg. conf.	Accuracy	Interpretation
0.5–0.7	5	0.60	0.60	roughly calibrated
0.7–0.8	5	0.76	0.40	overconfident
0.8–0.9	6	0.84	0.67	overconfident
0.9–1.0	4	0.93	0.50	strongly overconfident

Takeaway: Calibration compares predicted confidence with empirical frequency.

Accuracy versus calibration

Question: Can a model be accurate but poorly calibrated?

Accuracy versus calibration: answer

Yes.

Accuracy only asks whether the final class prediction was correct.

Calibration asks whether predicted probabilities match observed frequencies. For example, cases assigned probability 0.8 should be correct about 80% of the time.

Brier score: how close were the probabilities?

For a binary outcome $y \in \{0, 1\}$ and predicted probability p :

$$\text{Brier score for one case} = (p - y)^2.$$

Average this over all cases:

$$\text{Brier score} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2.$$

Case	p	y	$(p - y)^2$
confident and correct	0.90	1	0.01
uncertain and correct	0.55	1	0.20
confident and wrong	0.90	0	0.81

Takeaway: Lower is better. Confident wrong probabilities are costly.

Using scikit-learn for calibration

In the exercise, scikit-learn is just the source of realistic model probabilities.

The script does four things:

1. creates a labelled classification problem;
2. trains a logistic regression model and random forest models;
3. asks each model for predicted probabilities using `predict_proba`;
4. recalibrates random-forest probabilities using sigmoid and isotonic calibration.

Takeaway: The point is not to learn every API call. The point is to get probability forecasts from real classifiers and ask whether they are trustworthy.

Reading the calibration lab output

The script prints several measures for each model.

Output	What it tells you
accuracy	how often the final label is correct
Brier score	squared error of predicted probabilities
log loss	penalty for confident wrong probabilities
ECE	confidence-frequency gap across bins
reliability plot	visual check against perfect calibration

Compare calibrated and uncalibrated models. A model can have similar accuracy but better probability quality after calibration.

Takeaway: For decisions under uncertainty, a useful model needs more than correct labels; it needs probabilities that mean what they say.

What you should be able to do

By the end of the tutorial, you should be able to:

- ▶ compute one posterior from a frequency table
- ▶ explain why $P(A \mid B)$ differs from $P(B \mid A)$
- ▶ identify whether a row is a belief, likelihood, or unnormalised product
- ▶ read a small Bayes net as a modelling claim
- ▶ explain one example of explaining away
- ▶ predict qualitatively how sensing and motion change belief
- ▶ explain why the most likely state is not always the right action
- ▶ explain overconfidence using calibration bins

Question: Which concept should we practise once more before leaving?